

The Business Problem (as Rohan, VP Collections & Risk)

Here's the brief again, consolidated in one place so you have it as your project's "problem statement":

"Our NPA (default) rate has risen from 3.2% to 5.8% over 18 months. I need to know which borrower and account segments are driving this, so my credit team can recalibrate our scorecard and my collections team can flag at-risk accounts earlier. I need this backed by data, not gut feeling — and I want a dashboard my team can check regularly, not a one-time report."

Your job: use the loan, account, client, and district data to answer **which segments default more, and what account behavior signals trouble early**.

The Key Questions to Analyze in SQL

A. Baseline & Overview

1. What's the overall default rate across all loans?

B. Segment Analysis — Where are defaults concentrated?

2. Does default rate vary by **district/region**? Which districts are worst?
3. Is there a relationship between **district-level unemployment rate / average salary** (from the `district` table) and default rate? (This is your "macro-economic risk" angle)
4. Does default rate vary by **loan duration**? (short-term vs. long-term)?
5. Does default rate vary by **loan amount**? (bucket into small/medium/large)
6. Does default rate vary by **monthly payment size** relative to the loan?
7. Does default rate vary by **loan amount** and **monthly payment size** relative to the loan?

C. Account & Client Behavior — Early warning signals

8. Do defaulted accounts have a **lower average balance** (from `trans`) compared to good accounts?
9. Do defaulted accounts show more **withdrawal activity** or irregular transaction patterns before the loan went bad?
10. Does the **account's age** (how long it existed before the loan was taken) affect default risk? (Newer accounts = riskier?)
11. Does having a **credit card** (`card` table) linked to the account correlate with better/worse repayment?

D. Client Demographics

12. Does client **age or gender** show any default pattern? (Use `client.birth_date` to calculate age at loan issue)

E. Time Trends

13. Has the default rate changed over time based on **loan issue date** (`loan.date`)? (Are recent loans riskier than older ones — mirrors Rohan's "is loan quality getting worse" concern)

